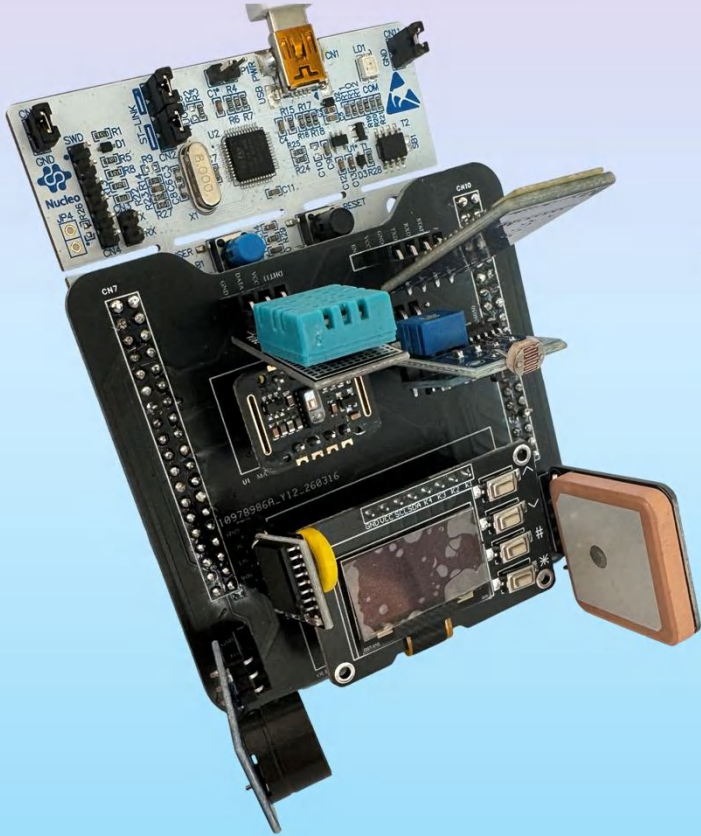


Choose Us! Hourglass!



Product Instruction
Product Instruction

Your Private Manager

Power On

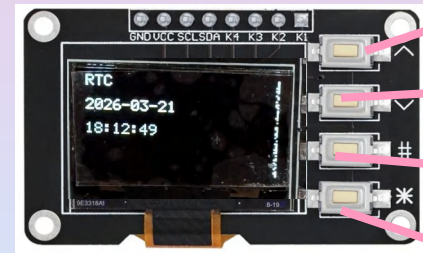
Switch page

Previous Page

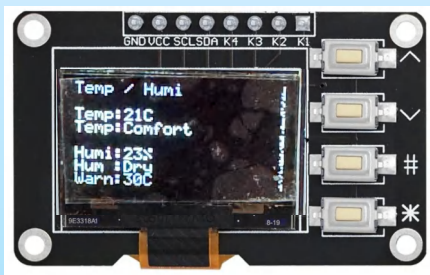
Next Page

Increase Alarm Threshold
Or reset step to 0

Decrease Alarm Threshold



Page 1:
Directly show real-time

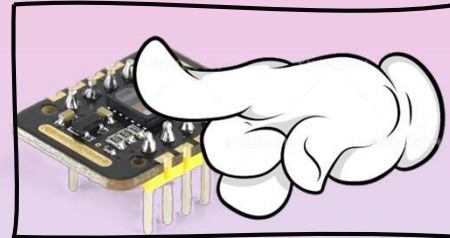


Page 2:
Directly show environment
Temperature and Humidity



Page 3:
Showing Heart Rate and
Blood Oxygen when
FINGER PUT ON MAX30102

NB. Buzzer will alarm when Heart
Rate over Threshold Setting.



1

2

3

4

Directly see position
information (OUTDOOR)

Position information sent
via Bluetooth (INDOOR)



Page 4:
Showing Position

Your Private Manager

How to send position information
via Bluetooth:

Step 1:
Open Serial Port Assistant

Step 2:
Select Port: COM6
Select Baud Rate: 9600
Select Data Bit: 8
Cheak Bit: None
Stop Bit: 1

端口名:	COM6
波特率:	9600
数据位:	8
校验位:	None
停止位:	1

接收宽度	<	>	使用说明	发送
BTGPS,39.7341,116.1698				

Step 4:
Check weather:
1. SRC change to: BT FIX
2. Position information



Step 3:
Entry:
BTGPS,Latitude,Longitude
Click: Send

5

6



Page 5:
Directly Illumination Intensity



Page 6:
Showing step count and motion

Reset step to 0

Your Private Manager

Access GUI

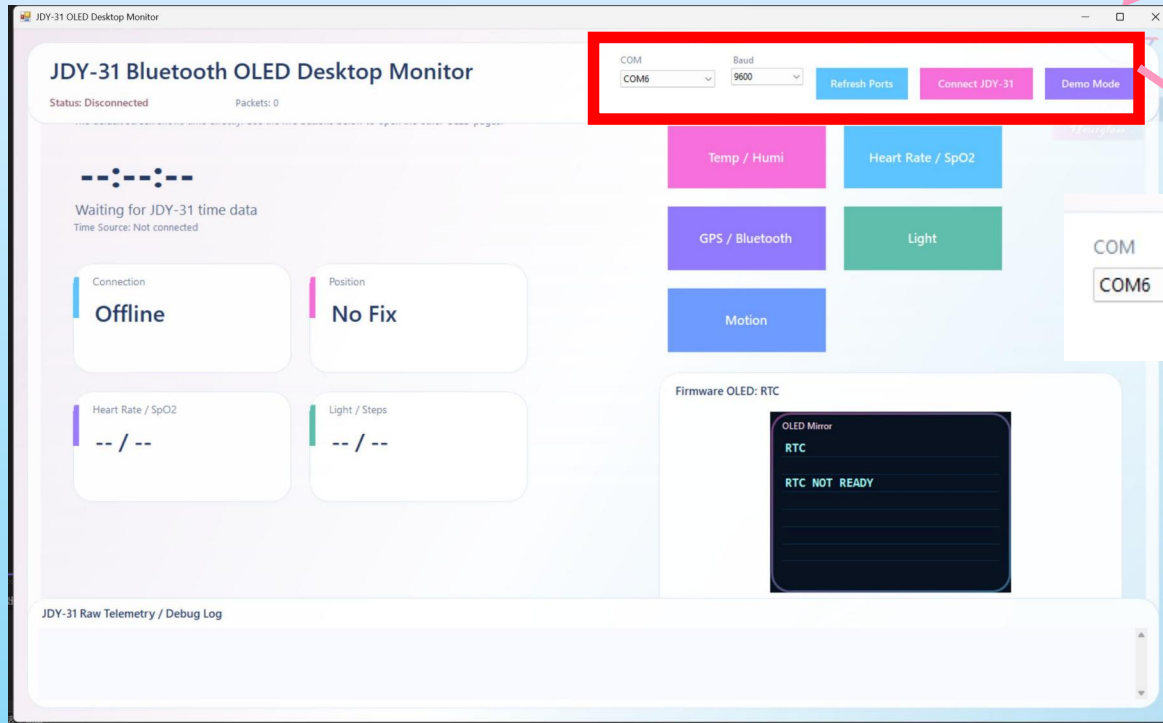
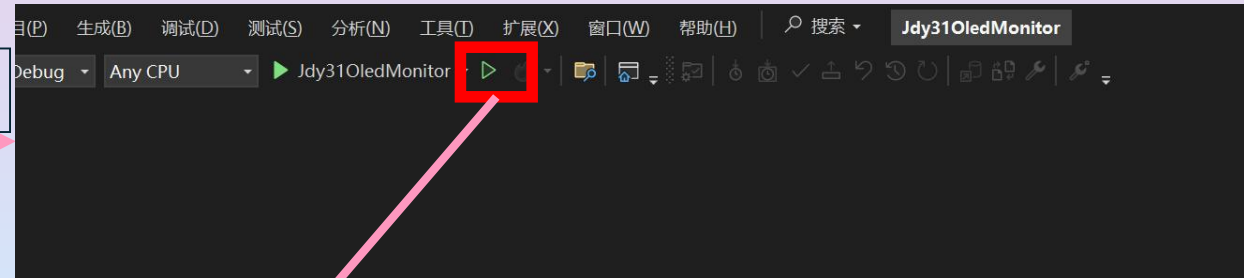


Go to:

EL2205 Coursework GP-3\Code for GUI\Jdy31OledMonitor\Jdy31OledMonitor.csproj

名称	修改日期	类型
Assets	2026/3/18 8:26	文件夹
bin	2026/3/18 8:27	文件夹
obj	2026/3/18 9:31	文件夹
BufferedFlowLayoutPanel.cs	2026/3/18 8:26	C# Source File
BufferedPanel.cs	2026/3/18 8:26	C# Source File
GradientPanel.cs	2026/3/18 8:26	C# Source File
Jdy31OledMonitor.csproj	2026/3/18 8:26	C# Project File
MainForm.cs	2026/3/18 8:26	C# Source File
OledPreviewControl.cs	2026/3/18 8:26	C# Source File
Program.cs	2026/3/18 8:26	C# Source File
RoundedPanel.cs	2026/3/18 8:26	C# Source File
TelemetryParser.cs	2026/3/18 8:26	C# Source File
TelemetryState.cs	2026/3/18 8:26	C# Source File

Open by
Visual Studio



1. Select Port: COM6
 2. Select Baud Rate: 9600
- Then, click “Connect JDY-31”

After Connect JDY-31



JDY-31 Bluetooth OLED Desktop Monitor

COM: COM6 Baud: 9600

Refresh Ports Disconnect Demo Mode

Status: JDY-31 connected Packets: 5

20:21:19
2026-03-21
Time Source: JDY-31 / DS3231

Temp / Humi Heart Rate / SpO2

GPS / Bluetooth Light

Motion

Connection: Online

Position: No Fix

Heart Rate / SpO2: -- / --

Light / Steps: 71% / 75

Firmware OLED: RTC

OLED Mirror

RTC

2026-03-21

20:21:19

JDY-31 Raw Telemetry / Debug Log

```
20:21:17 TIME=2026-03-21 20:21:15,TEMP=20,HUMI=25,COMFORT=Dry,HR=--,SPO2=--,FINGER=OFF,STEP=75,LIGHT=70,ADC_RAW=1229,ROLL=-175,PITCH=-42,YAW=0,AX=12732,AY=-1016,AZ=-14042,GX=187,GY=-52,GZ=-22,TEMP_WARN=30,H
20:21:18 TIME=2026-03-21 20:21:16,TEMP=20,HUMI=25,COMFORT=Dry,HR=--,SPO2=--,FINGER=OFF,STEP=75,LIGHT=71,ADC_RAW=1204,ROLL=-175,PITCH=-42,YAW=0,AX=12748,AY=-1024,AZ=-13984,GX=188,GY=-50,GZ=-22,TEMP_WARN=30,H
20:21:19 TIME=2026-03-21 20:21:17,TEMP=20,HUMI=25,COMFORT=Dry,HR=--,SPO2=--,FINGER=OFF,STEP=75,LIGHT=71,ADC_RAW=1199,ROLL=-175,PITCH=-42,YAW=0,AX=12686,AY=-1044,AZ=-13996,GX=188,GY=-52,GZ=-22,TEMP_WARN=30,H
20:21:20 TIME=2026-03-21 20:21:18,TEMP=20,HUMI=25,COMFORT=Dry,HR=--,SPO2=--,FINGER=OFF,STEP=75,LIGHT=71,ADC_RAW=1199,ROLL=-175,PITCH=-42,YAW=0,AX=12756,AY=-1006,AZ=-13950,GX=187,GY=-51,GZ=-22,TEMP_WARN=30,H
20:21:21 TIME=2026-03-21 20:21:19,TEMP=20,HUMI=25,COMFORT=Dry,HR=--,SPO2=--,FINGER=OFF,STEP=75,LIGHT=71,ADC_RAW=1199,ROLL=-175,PITCH=-42,YAW=0,AX=12702,AY=-1008,AZ=-13996,GX=187,GY=-52,GZ=-22,TEMP_WARN=30,H
```

OLED Page 1

OLED Page 2

OLED Page 4

OLED Page 6

OLED Page 3

OLED Page 5

OLED Mirror

Your Private Manager

EMERGENCY

If you encounter any below situation, please press the BLACK button to reset system.

1. Pressed OLED button but no reaction
2. Any changeful data has no change for a long time.
3. Removed any modules
4. Inserted any modules
5. Bluetooth cannot communication



University of
Lancashire



**Thanks for You
Choosing Our Product**

Product Name:	Hourglass
Developer:	DUAN Yuxin & XIAO Kewen
Contract:	YDuan8@lancashire.ac.uk
	KXiao1@lancashire.ac.uk
University of Lancashire, UK Beijing Institute of Technology EL2205-Assessment-Product Instruction Academic Year:2025-2026	

Your Private Manager